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Inventor(s)	Whitta; Jacob H. et al.

Shelf assembly for high capacity storage

Abstract

A shelf system includes at least one modular shelf assembly including at least one shelf unit, a plurality of support posts, an adjustable foot, covers, and anchors. The at least one shelf unit, plurality of support posts, adjustable foot, covers, and anchors are, in combination, adapted to create the modular shelf assembly. Each shelf unit is adapted to receive the plurality of support posts, including one support post generally located adjacent a rear side of the shelf unit about half-way between a first and a second end of the shelf unit.

Inventors: Whitta; Jacob H. (Bluffton, OH), Sayalith; Ben (Findlay, OH), Hummel; Eric (Findlay, OH), Vilay; Tony (Findlay, OH)

Applicant: CREATIVE PLASTIC CONCEPTS, LLC (Sycamore, OH)

Family ID: 86056897

Assignee: CREATIVE PLASTIC CONCEPTS, LLC (Sycamore, OH)

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Primary Examiner: Rodden; Joshua E

Attorney, Agent or Firm: Thompson Coburn LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. application Ser. No. 17/974,090, filed on Oct. 26, 2022 which claims the benefit of U.S. Provisional Application Ser. No. 63/271,772, filed on Oct. 26, 2021. The disclosure of the above application is incorporated herein by reference.

FIELD

(1) The present disclosure relates to shelf assemblies, and more particularly, to shelves and support posts for shelf assemblies.

INTRODUCTION

(2) The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

(3) Modular shelving systems include a plurality of shelves vertically stacked and interconnected via vertical support posts. Typically, each shelf includes an opening at each corner configured to receive an end of a support post thereby allowing successive shelves to be stacked by inserting support posts in the openings of adjacent shelves. The number of shelves and height may be influenced by what the shelves are being used for and the amount of space available where the shelf system is placed.

(4) Undesirably, when a shelf is heavily loaded, the shelf may deflect or sag wherein the shelf is no longer a substantially flat surface and thereby reducing the structural integrity of not only the shelf, but the entire shelving system. This can lead to the items placed on the shelf system to shift, tip over, and/or fall causing damage to the items and anything else surrounding the shelving system. Heavy loads can also cause the support posts to bow in such a way that further reduces the structural integrity resulting in the shelving system itself tipping over and possibly causing damage to the items on the shelf, the shelving system and anything else in the vicinity of the shelving system.

(5) There is a continuing need for a shelving system with increased structural integrity that can properly support various loads.

SUMMARY

(6) In concordance with the instant disclosure, a shelving system with increased structural integrity that can properly support various loads, is surprisingly discovered.

(7) In an exemplary embodiment, a shelf system including five support posts, which provides greater structural integrity for properly supporting various loads, and which reduces deflection and/or sag at the center of a shelf.

(8) In one embodiment, a shelf unit for a shelf assembly, comprises a shelf body and a plurality of support posts. The shelf body has a first end, a second end, a front side, a rear side, and an integral receiver. The integral receiver of the shelf body is disposed adjacent the rear side of the shelf body between the first end and the second end of the shelf body. One of the plurality of support posts is an integral receiver post received by the integral receiver of the shelf body.

(9) In another embodiment, a modular shelf assembly comprises a shelf unit, a plurality of support posts and at least one anchor. The shelf unit includes a shelf body and at least one shelf end cap.

The shelf body has a first end, a second end, a front side, a rear side, at least one connecting arm opening, at least one upper support opening and at least one lower support opening, and an integral receiver. The integral receiver of the shelf body is disposed adjacent the rear side of the shelf body between the first end and second end of the shelf body. The integral receiver is reinforced by at least one reinforcing member integrally connected to the integral receiver and an adjacent wall. The at least one shelf end cap has at least one connecting arm with at least one end cap receiver formed in the at least one connecting arm. The at least one connecting arm is received in the at least one connecting arm opening in the shelf body. The at least one end cap receiver is in substantial axial alignment with the at least one upper support opening and the at least one lower support opening of the shelf body. One of the plurality of support posts is an integral receiver post received by the integral receiver of the shelf body. At least one of the plurality of support posts is a corner post received by the at least one end cap receiver. The at least one anchor cooperates with at least one of the plurality of support posts to facilitate securing the shelf unit to an adjacent structure.

(10) In yet another embodiment, a kit for a shelf assembly, comprises a shelf unit, a plurality of support posts, at least one anchor, a plurality of covers and a plurality of feet. The shelf unit including a shelf body and at least one shelf end cap. The shelf body has a first end, a second end, a front side, a rear side, at least one connecting arm opening, at least one upper support opening and at least one lower support opening, and an integral receiver. The integral receiver is disposed adjacent the rear side of the shelf body between the first end and second end of the shelf body and the integral receiver is reinforced by at least one reinforcing member integrally connected to the integral receiver and an adjacent wall. The at least one shelf end cap has at least one connecting arm with at least one end cap receiver formed in the at least one connecting arm. The at least one connecting arm is configured to be received in the at least one connecting arm opening in the shelf body and position the at least one end cap receiver in substantial axial alignment with the at least one upper support opening and the at least one lower support opening of the shelf body. One of the plurality of support posts is an integral receiver post configured to be received by the integral receiver of the shelf body. At least one of the plurality of support posts is a corner post configured to be received by the at least one end cap receiver. The at least one anchor is configured to cooperate with at least one of the plurality of support posts to facilitate securing the shelf unit to an adjacent structure. Each of the plurality of covers is configured to be received by one of the integral receiver of the shelf body and the at least one end cap receiver of the at least one shelf end cap. Each of the plurality of feet is configured to be received by one of the integral receiver of the shelf body and the at least one end cap receiver of the at least one shelf end cap.

(11) a modular shelf assembly, comprises a shelf unit including a shelf body and at least one shelf end cap, the shelf body having a first end, a second end, a front side, a rear side, at least one connecting arm opening, at least one upper support opening and at least one lower support opening, and an integral receiver, wherein the integral receiver is adjacent the rear side of the shelf body between the first end and second end of the shelf body and the integral receiver is reinforced by at least one reinforcing member integrally connected to the integral receiver and an adjacent wall, the at least one shelf end cap having at least one connecting arm with at least one end cap receiver formed in the at least one connecting arm, the at least one connecting arm configured to be received in the at least one connecting arm opening in the shelf body and position the at least one end cap receiver in substantial axial alignment with the at least one upper support opening and the at least one lower support opening of the shelf body; a plurality of support posts wherein one of the plurality of support posts is an integral receiver post configured to be received by the integral receiver of the shelf body and at least one of the plurality of support posts is a corner post configured to be received by the at least one end cap receiver; and at least one anchor, wherein the at least one anchor is configured to cooperate with at least one of the plurality of support posts to facilitate securing the shelf unit to an adjacent structure.

(12) In a further embodiment, a kit for a shelf assembly, comprises a shelf unit including a shelf

body and at least one shelf end cap, the shelf body having a first end, a second end, a front side, a rear side, at least one connecting arm opening, at least one upper support opening and at least one lower support opening, and an integral receiver, wherein the integral receiver is adjacent the rear side of the shelf body between the first end and second end of the shelf body and the integral receiver is reinforced by at least one reinforcing member integrally connected to the integral receiver and an adjacent wall, the at least one shelf end cap having at least one connecting arm with at least one end cap receiver formed in the at least one connecting arm, the at least one connecting arm configured to be received in the at least one connecting arm opening in the shelf body and position the at least one end cap receiver in substantial axial alignment with the at least one upper support opening and the at least one lower support opening of the shelf body; a plurality of support posts wherein one of the plurality of support posts is an integral receiver post configured to be received by the integral receiver of the shelf body and at least one of the plurality of support posts is a corner post configured to be received by the at least one end cap receiver; at least one anchor, wherein the at least one anchor is configured to cooperate with at least one of the plurality of support posts to facilitate securing the shelf unit to an adjacent structure; a plurality of covers, wherein each of the plurality of covers is configured to be received by one of the integral receiver of the shelf body and the at least one end cap receiver of the at least one shelf end cap; and a plurality of feet, wherein each of the plurality of feet is configured to be received by one of the integral receiver of the shelf body and the at least one end cap receiver of the at least one shelf end cap.

(13) In a particular embodiment, a modular shelf system having a plurality of shelf assemblies vertically stacked in alignment with one another is shown. The shelf assemblies are stacked one over the other. A plurality of support posts extend between each shelf assembly. In one example, the shelf system includes five shelf assemblies and five support post extending between each shelf assembly configured to interconnect and stabilize the shelf assemblies.

(14) In a non-limiting example, the shelf system may be formed from a lightweight plastic material such as a polypropylene plastic material, by a molding process such as injection molding. However, it should be understood that the shelf system may be formed from any suitable material, including other thermoplastic materials such as polyethylene, and non-thermoplastic materials such as foam or metal. Nonetheless, one skilled in the art may employ any suitable process for forming the shelf system, as desired.

(15) Each shelf assembly includes a shelf body having a first end and a second end, and a shelf end cap connected to each one of the first end and the second end. The shelf end cap is configured to provide additional stability to the shelf system. In a non-limiting example, the shelf assemblies may be substantially rectangular, however, it should be understood that one skilled in the art may employ other shapes, as desired.

(16) Each shelf body further includes a top side, a bottom side, and a plurality of receiving holes extending between the top side and bottom side of the shelf body. Each shelf body includes five receiving holes, wherein four of the five receiving holes are located at corresponding corners of the shelf body and the fifth receiving hole is centrally located at a rear portion of the shelf body. More specifically, the shelf body includes a first receiving hole at a front right corner, a second receiving hole at a front left corner, a third receiving hole at a rear right corner, a fourth receiving hole at a rear left corner, and a fifth receiving hole between the third and fourth receiving holes. The third receiving hole, fourth receiving hole, and fifth receiving hole may be equally spaced along the rear portion of the shelf body.

(17) Each receiving hole includes an upper opening on the top side of the shelf body and a lower opening on the bottom side of the shelf body. The upper and lower openings are configured to facilitate stacking of the shelf assemblies. It should be appreciated that one skilled in the art may employ any number the receiving holes or the location of the receiving holes depending on the overall shape of the shelf body.

(18) Each support post includes a top portion and a bottom portion. The top portion of the support post is configured for sliding engagement with the lower opening of the receiving hole. The bottom portion of the support post is configured for sliding engagement with the upper opening of the receiving hole. As such, the upper opening and lower opening of each receiving hole is geometrically shaped to correspond with a shape of the top portion and bottom portion of the support posts, respectively. For example, the lower opening of the receiving hole is square shaped to correspond with a support post defining a square shape.

(19) The top portion of the support post may define a width less than a width of the lower opening of the shelf body to facilitate fitting of the top portion into the lower opening. The bottom portion of the support post may define a width less than a width of the upper opening of the shelf body to facilitate fitting of the bottom portion into the upper opening.

(20) The shelf system may further include a top cap configured to be disposed through the upper opening of each receiving hole on the upper most shelf assembly. The top cap includes a cap body defining a width that is smaller than a width of the upper opening of the shelf body to facilitate fitting of the cap body into the upper opening.

(21) The shelf end cap includes at least two arms configured to be inserted into arm openings on the first and second ends of each shelf body. Each arm is shaped such that an interference fit is established between each arm and the corresponding arm opening. Each arm of the shelf end cap further defines an aperture having a geographic shape corresponding to the shape of the support post and receiving holes. Each aperture is square shaped to correspond to the square shaped support posts and receiving holes.

(22) In operation, the two arms of the shelf end cap are inserted into arm openings in the ends of the shelf body. Once the shelf end cap is installed, the aperture of the shelf end cap is aligned with the receiving hole of the shelf body. Next, the top cap is inserted into the upper opening on the top side of the shelf body. When the top cap is pushed into place, the cap body is in mating contact with inner walls of the aperture of the shelf end cap and an inner wall of the receiving hole establishing an interference fit.

(23) In certain embodiments, each shelf end cap may have a plurality of interlocking apertures formed on an outer wall, a first side wall, and a second side wall of each shelf end cap. In a non-limiting example, there may be two interlocking apertures formed in the outer wall, and a single interlocking aperture formed in each side wall. The interlocking apertures may be configured to receive an interlocking component. The interlocking component may have a symmetrical shape, which permits the interlocking component to be received by different interlocking apertures of the same shape.

(24) In one embodiment, the interlocking component is generally H-shaped. In this embodiment, for example, the interlocking apertures are generally T-shaped to receive a portion of the H-shaped interlocking component. In particular, a T-shaped portion of the at least one interlocking component is configured to be received by the T-shaped interlocking aperture. However, one of ordinary skill in the art may select other suitable shapes for the interlocking aperture and the interlocking component, as desired. Advantageously, the interlocking apertures and the interlocking component are used to connect multiple shelf assemblies. For example, a first connecting portion of the interlocking component is disposed in each shelf end cap of a first shelf assembly, and a second connecting portion of the interlocking component is disposed on the shelf end cap of a second shelf assembly.

(25) The upper most (top) shelf assembly has a top cap installed in the upper opening of each one of the five receiving holes.

(26) The shelf system may include an adjustable foot and an adjustor adapted to reinforce the shelf system. The adjustable foot may be disposed within the lower opening of the receiving holes of the bottom most shelf assembly of the shelf system. The adjustable foot is configured to alter the height of the shelf system. The adjustable foot may have an externally threaded portion and the

adjustor may have an internally threaded portion, whereby the height of the adjustable foot is adjusted by rotating the adjustable foot clockwise or counterclockwise. For example, the adjustor may lower the height of the adjustable foot by rotating clockwise and raise the height of the adjustable foot by rotating counterclockwise.

(27) The shelf system may include a wall anchor configured to secure the shelf system to an adjacent wall thereby inhibiting the shelf system from tipping over. The wall anchor includes a front wall and opposing side walls configured to engage a support post adjacent the wall. Each opposing side wall includes an outwardly extending protrusion defining a through hole for receiving an attachment feature, such as a nail or screw, for attaching the wall anchor to the wall. When the wall anchor is installed, the front wall, opposing side walls, and the wall enclose the support post thereby securing the shelf system to the wall. One skilled in the art may employ two or any number of wall anchors, as desired.

(28) The shelf system may include a plurality of shelf assemblies, where the shelf assemblies are vertically stacked one over the other in alignment with one another. A plurality of support posts extend between each shelf assembly and are configured to interconnect and stabilize the shelf assemblies. Each shelf assembly includes a shelf body, a shelf end cap attached to ends of the shelf body, a plurality of receiving holes, and a plurality of top caps disposed in a corresponding receiving hole. The plurality of receiving holes are disposed at each corner of the shelf body. In a non-limiting example, a fifth receiving hole is centrally located at the rear portion of the shelf body. The difference between this shelf system and other embodiments is that each receiving hole defines an octagonal shape, as opposed to a square shape.

(29) The octagonal receiving holes are configured to receive an octagonal support post to facilitate stacking of the shelf assemblies. The support post and receiving hole connection further provides stability and support to the shelf system. It has been advantageously found that providing a fifth support post, centrally located between opposing corners, inhibits the center of the shelf body from sinking when the shelf is heavily loaded thereby increasing the structural integrity of the shelf system.

(30) It should be understood that the plurality of top caps define a shape that corresponds to the shape of the receiving hole.

(31) A modular shelf system according to another embodiment of the present disclosure includes that each receiving hole of this shelf system defines a circular shape instead of square or octagon shapes. Each circular receiving hole is configured to receive a circular support post. Furthermore, the shelf system may include a plurality of top caps defining a shape corresponding to the shape of the receiving hole.

(32) It has been advantageously found that providing a fifth support post, centrally located between opposing corners, inhibits the center of the shelf body from sinking when the shelf is heavily loaded thereby increasing the structural integrity of the shelf system.

(33) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

(2) FIG. 1 is a top perspective view of a modular shelf assembly for a high capacity storage shelf according to one embodiment of the present disclosure;

(3) FIG. 2 is a top perspective view of two modular shelf assemblies joined by interlocking

components;

(4) FIG. 3 is an exploded top perspective view of the shelf assembly according to another embodiment of the present disclosure;

(5) FIG. 4 is a top perspective view of a shelf unit of the modular shelf assembly shown in FIG. 1 according to another embodiment of the present disclosure;

(6) FIG. 5 is a bottom perspective view of the shelf unit shown in FIG. 4;

(7) FIG. 6 is a cross-sectional side elevational view of the shelf unit taken at section line A-A in FIG. 4;

(8) FIG. 7 is an enlarged, fragmentary bottom perspective view of the shelf unit taken at call-out B in FIG. 5, showing an integral receiver and integral receiver support, a plurality of crossbars and reinforcing bars, and reinforcing strips of the shelf unit;

(9) FIG. 8 is an enlarged, fragmentary top perspective view of a shelf unit taken at call-out C in FIG. 4, showing the substantially square shaped integral receiver;

(10) FIG. 9 is an enlarged, fragmentary top perspective view of a shelf unit taken at call-out C in FIG. 4 according to another embodiment of the present disclosure, illustrating a shelf unit having an octagonal shaped integral receiver; and

(11) FIG. 10 is an enlarged, fragmentary top perspective view of a shelf unit taken at call-out C in FIG. 4 according to another embodiment of the present disclosure, illustrating a shelf unit having a round shaped integral receiver.

DETAILED DESCRIPTION

(12) The following description of technology is merely exemplary in nature of the subject matter, and is not intended to limit the present disclosure, application or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

(13) Modular shelf assemblies are disclosed in applicant's co-owned U.S. Pat. No. 11,350,743, the entire disclosure of which, is incorporated herein by reference. Where any conflict or ambiguity can exist between U.S. Pat. No. 11,350,743 and this detailed description, the present detailed description controls.

(14) Referring to FIGS. 1-10, a modular shelf assembly **1** is shown that is adapted to interlock with adjacent modular shelf assemblies **1**. In certain embodiments, the modular shelf assembly **1** may include at least one shelf unit **100**, a plurality of support posts **200**, an adjustable foot **300**, and an anchor **400**. The at least one shelf unit **100**, the plurality of support posts **200**, the adjustable foot **300**, and the anchor **400**, in combination, are adapted to create the modular shelf assembly **1**.

(15) In a non-limiting example, the shelf assembly **1** may be formed from a lightweight plastic material such as a polypropylene plastic material, by a molding process such as injection molding. However, it should be understood that the shelf assembly **1** may be formed from any suitable material or a plurality of suitable materials, including other thermoplastic materials such as polyethylene, and non-thermoplastic materials such as foam or metal. Nonetheless, one skilled in the art may employ any suitable process for forming the shelf system **2**, as desired.

(16) With further reference to FIGS. 1-7, and in particular embodiments, each shelf unit **100** may include a shelf body **102** having a top side **104**, a bottom side **106**, a front side **108**, a rear side **110**, a first end **112**, a second end **114**, and at least one shelf end cap **150**. The at least one shelf end cap **150** may include a first shelf end cap **150** and a second shelf end cap **150**. The first shelf end cap **150** may be adapted to connect with the first end **112** of the shelf body **102**, and the second shelf end cap **150** may be adapted to connect with the second end **114** of the shelf body **102**, thereby creating the shelf unit **100**.

(17) In a non-limiting example, the shelf unit **100** may be substantially rectangular and have four corners. However, other shapes for the shelf unit **100** may also be selected by a skilled artisan, within the scope of the present disclosure.

(18) With further reference to FIGS. 3, in certain embodiments, the shelf body **102** may have a

mostly hollow interior that includes connecting arm openings **118** formed in the first end **112** and the second end **114** of the shelf body **102**. The connecting arm openings **118** may be adapted to receive a connecting arm **152** disposed on the at least one shelf end cap **150**. Support members **120** are provided adjacent the first end **112** and the second end **114** of the shelf body **102**. In the illustrated embodiment, the support members **120** extend between and are integrally formed with the top side **104** and the bottom side **106** of the shelf body **102**.

(19) As shown in FIGS. 5-7, the shelf body **102** may include at least one crossbar **122** that extends from the first end **112** to the second end **114** of the shelf body **102**. The at least one crossbar **122** may be hollow and integrally molded in the shelf body **102** and adapted to support and minimize a deflection of the top side **104** of the shelf body **102**. In the illustrated embodiment, three of the at least one crossbars **122** are provided, a front crossbar **122a** adjacent the front side **108** of the shelf body **102**, a rear crossbar **122c** adjacent the rear side **110** of the shelf body **102**, and a middle crossbar **122b** substantially centered between and parallel to the front side **108** the rear side **110** of the shelf body **102**. The shelf body **102** may also include at least one reinforcing bar **124** that is integrally connected to at least one of the at least one crossbars **122**. In the illustrated embodiment, two of the at least one reinforcing bars **124** are provided, a front reinforcing bar **124a** and a rear reinforcing bar **124b**. The front reinforcing bar **124a** is disposed between and connected to the front crossbar **122a** and the middle crossbar **122b**. The rear reinforcing bar **124b** is disposed between and is connected to the middle crossbar **122b** and the rear crossbar **122c**. The at least one reinforcing bar **124** may include a plurality of reinforcing members **126** integrally formed therewith, wherein the at least one reinforcing bar **124** and reinforcing members **126** are configured to cooperate with the at least one crossbar **122** to support and minimize a deflection of the top side **104** of the shelf body **102**. Furthermore, one skilled in the art may select, as desired, other shapes, locations and number of the at least one crossbar **122** and the at least one reinforcing bar **124** as well as other shapes and locations of the reinforcing members **126** of the at least one reinforcing bar **124**.

(20) With continuing reference to FIGS. 5-7, in particular embodiments, the shelf body **102** may have a plurality of reinforcing strips **128**, where the reinforcing strips **128** are spaced apart from other of the reinforcing strips **128**, forming a gap **130** therebetween. The reinforcing strips **128** are connected to adjacent at least one crossbars **122** to cooperate therewith to support and minimize a deflection of the top side **104** of the shelf body **102**. As shown, through holes **132** may be formed in the top side **104** of the shelf body **102** in the area of the gap **130** between adjacent reinforcing strips **128**. It should be understood that the at least one crossbar **122**, the at least one reinforcing bar **124**, the reinforcing strips **128** and the through holes **132** cooperate to minimize a weigh of the shelf body **102** while maintain a desired strength and maximizing the weigh carry capacity of the modular shelf assembly **1**.

(21) With reference to FIGS. 3-7, each shelf body **102** may include a plurality of upper support openings **134** formed therein. Each of the upper support openings **134** is adapted to receive the support posts **200**. In the example shown, one of the upper support openings **134** is provided adjacent each corner of the shelf body **102** and is adapted to receive a corner post **208**.

(22) With reference to FIGS. 5, each shelf body **102** may have a plurality of lower support openings **136** formed therein. Each of the lower support openings **136** is adapted to receive the support posts **200**. Furthermore, each lower support opening **136** is aligned with one of the plurality of the upper support openings **134** to facilitate a stacking of support posts **200**. Accordingly, the lower support openings **136** may be disposed adjacent the corners of the shelf body **102** and adapted to receive the corner post **208**.

(23) Referring now to FIGS. 3-8, an integral receiver **138** is provided adjacent the rear side **110** of the shelf body **102** in the rear crossbar **122c**. The integral receiver **138** is adjacent and connected to an end of the rear reinforcing bar **124b** and configured to receive an integral receiver post **206**. In certain embodiments, the integral receiver **138** has dimensions that enable the integral receiver post **206** to be received in and secured by the integral receiver **138**. The integral receiver **138** is at least

partially surrounded by a reinforcing wall **141** that is spaced apart from the integral receiver **138**. A plurality of reinforcing ribs **140** are provided that extend from the integral receiver **138** to the reinforcing wall **141**, wherein the reinforcing ribs **140** and the reinforcing wall **141** cooperate to strengthen the integral receiver **138** and facilitate securely maintaining the integral receiver post **206** in the integral receiver **138**.

(24) With reference to FIGS. 3-5, the at least one shelf end cap **150** includes at least one of the connecting arms **152**. In the illustrated embodiment, one of the connecting arms **152** is provided adjacent each end of the at least one shelf end cap **150**. The connecting arm **152** is configured to be inserted into the connecting arm opening **118** of the shelf body **102**. The connecting arm **152** may be shaped such that an interference fit is established between the connecting arm **152** and the corresponding connecting arm opening **118**. The connecting arm **152** may include a locking member **154** adapted to cooperate with the shelf body **102** to facilitate the retention of the connecting arm **152** in the connecting arm opening **118**. The at least one shelf end cap **150** may also include one or more additional connecting members formed at selected locations along the length of the at least one shelf end cap **150**. In the illustrated embodiment, one such additional connecting member is a center connecting member **156** that is provided along with a center connecting member opening **119** in the shelf body **102**, wherein the center connecting member **156** is configured to be received by the center connecting member opening **119**. In the illustrated embodiment, the center connecting member **156** and the center connecting member opening **119** are configured in a similar shape and size as the connecting arm **152** and connecting arm opening **118** of the shelf body **102**, respectively. It should be understood that the center connecting member **156** and the center connecting member opening **119** may be configured to have a shape and size, such as a tab, for example, that is different from the connecting arm **152** and the connecting arm opening **118**, respectively.

(25) A plurality of tabs **158** are formed in the at least one shelf end cap **150** to facilitate the aligning of the at least one shelf end cap **150** with the shelf body **102** and to maximize a strength of the at least one shelf end cap **150**, wherein the tabs **158** may cooperate with the support members **120** of the shelf body **102** to further facilitate the aligning of the at least one shelf end cap **150** with the shelf body **102**.

(26) Each of the connecting arms **152** provided adjacent to the ends of the at least one shelf end cap **150**, include an end cap receiver **160** formed therein. When the connecting arms **152** are inserted into the connecting arm openings **118** of the shelf body **102**, the end cap receiver **160** is in substantial axial alignment with the upper support openings **134** and the lower support openings **136** to enable the support post **200** to pass through one of the upper support openings **134** or one of the lower support openings **136** and into the end cap receiver **160**. In the illustrated embodiment, the end cap receivers **160** are adapted to receive the corner post **208**. The end cap receiver **160** has dimensions that enable the corner post **208** to be received in and secured by the end cap receiver **160**. The at least one shelf end cap **150** is configured to provide additional stability to the shelf body **102**, the corner post **208**, and the entirety of the modular shelf assembly **1**.

(27) In certain embodiments, each at least one shelf end cap **150** may have at least one interlocking aperture **162** formed at a peripheral edge thereof. The at least one interlocking apertures **162** may be configured to receive an interlocking component **164**. The interlocking component **164** may have a symmetrical shape, which permits the interlocking component **164** to be jointly received by two of the at least one interlocking apertures **162** of the same shape. As shown in FIGS. 1-3, in one embodiment, the interlocking component **164** is generally H-shaped and the at least one interlocking apertures **162** are generally T-shaped to receive a portion of the H-shaped interlocking component **164**. Advantageously, the at least one interlocking apertures **162** and the interlocking component **164** are used to connect multiple shelf assemblies **1** as shown in FIG. 2.

(28) With further reference to FIGS. 1-3, each of the plurality of support posts **200**, includes a first end **202** and a second end **204** that are configured for sliding frictional engagement with the end

cap receiver **160** of the at least one shelf end cap **150** and the integral receiver **138** of the shelf body **102**. The first end **202** of the plurality of support posts **200** may define a width less than a width of the lower support opening **136** of the shelf body **102** to facilitate fitting the first end **202** through the lower support opening **136** of the shelf body **102** into the end cap receiver **160** of the at least one shelf end cap **150**. The second end **204** of the plurality of support posts **200** may define a width less than a width of the upper support opening **134** of the shelf body **102** to facilitate fitting the second end **204** through the upper support opening **134** of the shelf body **102** into the end cap receiver **160** of the at least one shelf end cap **150**.

(29) In certain embodiments, the first end **202** of one of the plurality of support posts **200** telescopically receives the second end **204** of another one of the plurality of support posts **200** to connect the two support posts, wherein the end cap receivers **160** of the at least one shelf end caps **150** and the integral receiver **138** of the shelf body **102** substantially surround the telescopic connection between two of the plurality of support posts **200** and maximize a strength thereof. In the illustrated embodiments, one of the plurality of support posts **200** is provided between shelf bodies **102**. It should be understood that two or more of the plurality of support posts **200** may be telescopically joined and disposed between two of the shelf bodies **102** in order to provide a desired distance between the shelf bodies **102**.

(30) The integral receiver post **206** received by the integral receiver **138** provided adjacent the rear side **110** of the shelf body **102** substantially increases the load carrying capability of the shelf unit **100** and the overall modular shelf assembly **1**. The increase in load carrying capacity is achieved while maximizing the clear span distance on the front side **108** of the shelf body **102**, which facilitates a user placing objects on the shelf body **102** of the modular shelf assembly **1**.

(31) The integral receiver post **206** cooperates with the integral receiver **138** and the integral receiver post **206** to minimize a deflection of the shelf body **102** when under a vertical load such as when objects are placed on the shelf body **102** of the modular shelf assembly **1**. Furthermore, the reinforcing ribs **140** and the reinforcing wall **141**, as well as the rear reinforcing bar **124b** structurally connected to the integral receiver **138** cooperate with the integral receiver post **206** to strengthen the shelf body **102** and the overall modular shelf assembly **1**. Further still, the integral receiver post **206** cooperates with the rear crossbar **122c**, the rear reinforcing bar **124b** and the reinforcing members **126**, and the integral receiver **138** and the plurality of reinforcing ribs **140** and the reinforcing wall **141** to maximize the strength of the shelf body **102** and the overall modular shelf assembly **1**.

(32) Referring to FIGS. **1-4**, the shelf unit **100** of the modular shelf assembly **1** may further include at least one cover **142**. The at least one cover **142** is adapted to be disposed through the upper support opening **134** and received by the end cap receiver **160** as well as be received by the integral receiver **138**. The at least one cover **142** are configured to telescopically attach to the first end **202** of the plurality of support posts **200** within the end cap receiver **160** and the integral receiver **138**. It should be appreciated that the at least one cover **142** militates against items falling into the integral receiver **138**, the end cap receivers **160**, and the plurality of support posts **200** of the modular shelf assembly **1**.

(33) Referring to FIGS. **1-3**, the modular shelf assembly **1** may include an adjustable foot **300** having a fixed member **302** and a moveable member **304**. With respect to the corner post **208**, the fixed member **302** is adapted to pass through the lower support opening **136** of the shelf body **102**, be received by an end cap receiver **160** of the at least one shelf end cap **150**, and telescopically connect to the second end **204** of the plurality of support posts **200**. With respect to the integral receiver post **206**, the fixed member **302** of the adjustable foot **300** is received by the integral receiver **138** formed in the shelf body **102** and telescopically connected to the second end **204** of the integral receiver post **206**. The moveable member **304** is adapted to be moveably connected to the fixed member **302**, wherein the moveable member **304** can be selectively moved with respect to the fixed member **302** to change an overall length of the adjustable foot **300** to facilitate a leveling

of the modular shelf assembly **1**. In the illustrated embodiment, the fixed member **302** and the moveable member **304** of the adjustable foot **300** are threaded to allow for relative rotational movement therebetween. It should be understood that other types of adjustable feet may be employed to facilitate changing the length of the foot and the leveling of the modular shelf assembly **1**.

(34) With continued reference to FIGS. **1-3**, the modular shelf assembly **1** may further include an anchor **400** configured to secure the modular shelf system **1** to an adjacent wall or other support structure. The anchor **400** includes a wall **402** with an outwardly extending protrusion **404** at each end thereof. Each outwardly extending protrusion **404** includes a through hole **406**. The anchor **400** is configured to engage at least a portion of the plurality of support posts **200** wherein the through hole **406** are adapted to receive an attachment feature, such as a nail or screw, for example, to facilitate attaching the anchor **400** to a wall or other support structure. The anchor **400** militates against the modular shelf system **1** from tipping over. It should be appreciated that while two of the anchors **400** are shown in FIGS. **1-3**, one skilled in the art may employ any number of the anchors **400**, as desired.

(35) As shown in FIGS. **1-8**, the plurality of support posts **200** as well as the upper support openings **134** and the lower support openings **136**, the integral receiver **138** of the shelf body **102**, and the end cap receivers **160** of the at least one shelf end cap **50** are substantially square shaped, wherein the corners thereof are radiused and/or chamfered. The cooperating generally square shapes facilitates an assembly of the modular shelf assembly **1**. It should be understood that other geometric shapes may be utilized as desired. For example, and as shown in FIGS. **8-9**, respectively, an octagonal and a round geometric shape are shown for the integral receiver **138**. It should also be understood that the plurality of support posts **200**, the upper support openings **134** and the lower support openings **136** and the integral receiver **138** of the shelf body **102**, the end cap receivers **160** of the at least one shelf end cap **150**, the at least one cover **142**, the adjustable foot **300**, and the anchor **400** may also have octagonal, round, or other cooperating geometric shapes to facilitate the assembly of the modular shelf assembly **1**.

(36) Additionally, in various embodiments, the shelf unit **100**, the plurality of support posts **200**, the at least one cover **142**, the adjustable foot **300**, the anchor **400**, and the interlocking component **164** may be sold separately as a kit. The kit enables the user to construct shelf assemblies **1** of varying heights and widths, as desired.

(37) While certain representative embodiments and details have been shown for purposes of illustrating the present disclosure, it will be apparent to those skilled in the art that various changes may be made without departing from the scope of the disclosure, which is further described in the following appended claims.

Claims

1. A shelf unit for a shelf assembly, comprising: a shelf body having a first end, a second end, a front side, a rear side, first and second support openings, an integral receiver, a first crossbar, a second crossbar, and a third crossbar, the first support opening disposed adjacent the first end and adjacent the rear side, the second support opening disposed adjacent the second end and adjacent the rear side, the integral receiver disposed adjacent the rear side between and spaced from the first end and the second end of the shelf body, the first crossbar being adjacent the front side and disposed between the first end and the second end, the second crossbar being adjacent the rear side and disposed between the first end and the second end, and the third crossbar being disposed between the first crossbar and the second crossbar and disposed between the first end and the second end, wherein the integral receiver is reinforced by at least one reinforcing rib integrally connected to the integral receiver and an adjacent wall; and a plurality of support posts, one of the plurality of support posts being an integral receiver post received by the integral receiver of the

shelf body.

2. The shelf unit of claim 1, wherein the shelf body has at least one reinforcing bar connected to the integral receiver and extending toward the front side of the shelf body.
 3. The shelf unit of claim 1, wherein the integral receiver is disposed in the second crossbar.
 4. The shelf unit of claim 1, wherein the shelf body has at least one reinforcing bar, and the at least one reinforcing bar is disposed between the first crossbar and the second crossbar.
 5. The shelf unit of claim 4, wherein the at least one reinforcing bar includes a plurality of reinforcing members.
 6. The shelf unit of claim 1, wherein the shelf body has a plurality of reinforcing strips, and each of the plurality of reinforcing strips is disposed between the first crossbar and the second crossbar.
 7. The shelf unit of claim 1, wherein the shelf body has at least two reinforcing bars, and a first reinforcing bar is disposed between the first crossbar and the third crossbar and a second reinforcing bar is disposed between the second crossbar and the third crossbar.
 8. The shelf unit of claim 7, wherein the integral receiver is disposed in the second crossbar and an end of the second reinforcing bar is connected to the integral receiver.
 9. The shelf unit of claim 1, wherein the shelf body has a plurality of reinforcing strips, and each of the plurality of reinforcing strips is disposed between the first crossbar and the third crossbar or the second crossbar and the third crossbar.
 10. The shelf unit of claim 1, wherein the shelf body includes at least one cover, and the at least one cover is inserted into the integral receiver of the shelf body.
 11. The shelf unit of claim 1, wherein the shelf body includes at least one foot, and the at least one foot is inserted into the integral receiver of the shelf body.
 12. The shelf unit of claim 1 including at least one anchor, wherein the at least one anchor cooperates with at least one of the plurality of support posts to facilitate securing the shelf unit to an adjacent structure.
 13. The shelf unit of claim 1, wherein the shelf body includes third and fourth support openings, the third support opening being at a front right corner of the shelf body, the fourth support opening being at a front left corner of the shelf body, wherein the integral receiver, the first support opening, and the second support opening are equally spaced along the rear side of the shelf body.
 14. A shelf unit for a shelf assembly, comprising: a shelf body having a first end, a second end, a front side, a rear side, first and second support openings, and an integral receiver, a first crossbar, and a second crossbar, the first support opening disposed adjacent the first end and adjacent the rear side, the second support opening disposed adjacent the second end and adjacent the rear side, the integral receiver disposed adjacent the rear side between and spaced from the first end and the second end of the shelf body, the first crossbar being adjacent the front side and disposed between the first end and the second end, the second crossbar being adjacent the rear side and disposed between the first end and the second end, wherein the integral receiver is reinforced by at least one reinforcing rib integrally connected to the integral receiver and an adjacent wall, and wherein the integral receiver is disposed in the second crossbar; and a plurality of support posts, one of the plurality of support posts being an integral receiver post received by the integral receiver of the shelf body.
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